RéseauFlux: Machine Learning-Guided Ranking of Metabolic Interventions for Succinate Overproduction in *Escherichia coli*

Abstract

Identifying optimal gene knockouts and overexpressions for metabolic engineering is computationally expensive when exhaustive flux balance analysis (FBA) is required for every candidate intervention. Here we present RéseauFlux, a machine learning pipeline that learns to rank metabolic interventions from a small enumerated training set and generalizes to the full genome-scale model without additional FBA calls. Applied to aerobic succinate production in *Escherichia coli* K-12 using the iJO1366 genome-scale model (2,583 reactions, 1,805 metabolites), the pipeline correctly identifies ATPS4rpp knockout as the top single intervention (16.38 mmol/gDW/h, consistent with prior computational predictions in the literature), achieves a holdout Spearman rank correlation of ρ = 0.602 on unseen reaction groups, and discovers a novel double intervention (ATPM KO + O2tex KO) yielding 16.80 mmol/gDW/h — 9% above the best single knockout. An ablation study demonstrates that gradient boosting regression (GBR) alone outperforms ensemble approaches incorporating a pairwise logistic ranker (ρ = 0.557 vs 0.358). The pipeline generalizes across three production targets (succinate, L-malate, acetate). All code is available at github.com/Daxlia/ReseauFlux.

Keywords: metabolic engineering, genome-scale model, machine learning, gradient boosting, succinate, flux balance analysis, strain design, *Escherichia coli*

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1. Introduction

Metabolic engineering aims to redirect intracellular flux toward a desired product by rationally modifying gene expression. Constraint-based reconstruction and analysis (COBRA) methods, particularly flux balance analysis (FBA), enable in silico prediction of metabolic phenotypes under genetic perturbations [1]. However, a genome-scale model such as iJO1366 contains 2,583 reactions, and evaluating all possible single, double, and triple knockouts via FBA scales combinatorially — a challenge that has motivated heuristic tools such as OptKnock [2], OptGene [3], and FastKnock [4].

Machine learning has been applied to accelerate strain design by learning flux response patterns from partial data [5,6]. Yet most approaches either require large experimental datasets, are limited to a narrow set of reactions, or do not rigorously quantify generalization to reaction types unseen during training — a critical distinction for assessing real-world utility.

We introduce RéseauFlux, a pipeline that (i) extracts reaction-level features combining flux variability analysis (FVA), network topology, and biochemical context; (ii) trains a gradient boosting regressor on rank-normalized production flux labels from approximately 300 enumerated interventions; and (iii) predicts intervention rankings across the full genome-scale model without further FBA. We report performance on aerobic succinate production as a primary target and validate generalization on L-malate and acetate.

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2. Methods

2.1 Genome-scale model and simulation conditions

The *E. coli* K-12 iJO1366 genome-scale model [7] was used throughout, loaded via COBRApy [8] with glucose minimal medium under aerobic conditions. The primary target was the succinate exchange reaction (EX_succ_e). Wild-type

(WT) reference fluxes were computed by maximizing the BIOMASS_Ec_iJO1366_core_53p95M objective.

Intervention phenotypes were evaluated using a two-step optimizer: (1) maximize mutant growth (g_max); (2) fix growth ≥ 10% of g_max and maximize EX_succ_e. This growth floor (GROWTH_FRACTION = 0.10) allows the optimizer to sacrifice up to 90% of growth capacity in exchange for production, generating a continuous label gradient across interventions. An intervention was deemed lethal if g_max < 10^{-6} h^{-1} and was assigned zero production flux.

Three intervention types were considered per reaction: knockout (KO, bounds set to [0, 0]), overexpression (OE, upper bound ×3 + 1), and downregulation (down, upper bound ×0.2).

2.2 Feature extraction

For each reaction in the model, 18 scalar features were computed, plus one-hot subsystem encoding (dimension varies with the model) and an intervention type indicator, yielding a total feature vector of approximately 35–40 dimensions for iJO1366. The 18 scalar features are organized into three groups:

FVA-derived features (7): WT flux (wt_flux); FVA minimum and maximum (fva_min, fva_max); FVA range (fva_range = fva_max − fva_min); FVA utilization ratio (fva_util = |wt_flux| / fva_range, capped at 10); FVA range normalized by WT growth rate (fva_range_per_g = fva_range / growth); and the flux/FVA saturation index (flux_fva_ratio = |wt_flux| / (fva_range + ε)).

Network topology features (7): reaction degree (number of metabolites involved); reversibility flag; betweenness centrality in the reaction-metabolite bipartite graph (NetworkX); shortest path distance to the target exchange reaction; shortest path distance to the biomass reaction; and mean and maximum shortest-path distances to other reactions.

Biochemical context features (4): flux entropy (Shannon entropy over metabolite stoichiometric weights); binary flags for whether the reaction produces or consumes the target metabolite; and the stoichiometric coefficient toward the target.

Categorical: one-hot subsystem membership across all subsystems in the model; integer encoding of intervention type (KO = 0, OE = 1, down = 2).

FVA was performed at 90% of WT growth to tighten flux bounds. All features were standardized (zero mean, unit variance) before model training.

2.3 Training label construction

The top 300 reactions ranked by FVA range (highest metabolic flexibility) were selected for enumeration. For each reaction and each intervention type, the two-step optimizer was applied, yielding a raw production flux y_flux. Labels were rank-normalized to remove the effect of the plateau distribution characteristic of genome-scale FBA:

$$y_\text{rank} = \frac{\text{rank}(y_\text{flux})}{N}$$

where N is the total number of enumerated samples (300 reactions × 3 intervention types = 900) and rank uses average ranking (scipy.stats.rankdata). This maps the plateau-dominated flux distribution to a uniform [0, 1] interval more suitable for gradient boosting regression.

2.4 Gradient boosting regression and cross-validation

A GradientBoostingRegressor (scikit-learn) was trained with the following hyperparameters: n_estimators = 300, max_depth = 3, learning_rate = 0.05, subsample = 0.8, min_samples_leaf = 2, random_state = 42. Performance was evaluated by 5-fold GroupKFold cross-validation with reactions as groups, preventing leakage between structurally similar reactions.

For the 300 enumerated reactions, exact ground-truth ranks (known_ranks) were used in the final ranking, bypassing GBR reconstruction. For the remaining 2,283 unenumerated reactions, GBR predictions were used directly.

2.5 Composite score

For FBA-validated interventions, a composite score was computed to balance production gain and growth viability:

$$\text{composite} = 0.85 \times \frac{t}{\text{max\_target}_{WT}} + 0.15 \times \frac{g}{g_{WT}}$$

where t is the production flux, g is the growth rate, and max_target_WT is the WT maximum production flux. Lethal interventions ($g < 10^{-6}$) are assigned composite = 0.

2.6 Ablation study

Three ranking strategies were compared by 5-fold GroupKFold Spearman ρ: GBR-only; pairwise logistic ranker (logistic regression on pairwise feature differences); and ensemble (0.6 × GBR score + 0.4 × pairwise win rate). The pairwise ranker was excluded from the final pipeline following the ablation results.

2.7 Holdout generalization test

Reaction groups were partitioned 70/30 by group identity (never by individual sample). The GBR was trained on 70% of reaction groups (210 reactions, 630 samples) and evaluated on the held-out 30% (90 reaction groups, 270 samples not seen during training). Reported metrics: Spearman ρ, Top-K precision (fraction of true top-K recovered among predicted top-K), and Top-5 precision (fraction of the model's five highest-flux reactions recovered in the five predicted top reactions).

2.8 Double intervention prediction

All pairwise combinations of the top-50 single predictions were scored using the arithmetic mean of individual predicted ranks, reflecting a no-epistasis prior:

$$\text{pred}_\text{pair} = \frac{\text{pred}_i + \text{pred}_j}{2}$$

The top 25 pairs by predicted score were validated with two-step FBA. A random baseline selected 25 pairs uniformly at random from the same candidate pool for comparison.

2.9 Monte Carlo robustness

For each top-10 validated single intervention, glucose and oxygen uptake bounds were independently perturbed across 15 trials using multiplicative Gaussian noise with σ = 20% (bound × (1 + 0.2 × N(0,1))). Robustness score = mean_flux − std_flux across trials.

2.10 Flux control coefficient analysis

Flux control coefficients (FCC) were computed numerically with ε = 5%. For each candidate reaction carrying non-zero WT flux v, the upper bound was set to v × (1 − ε) — constraining the reaction to 95% of its WT flux — and the resulting change in EX_succ_e flux was recorded relative to the reference production flux. Reactions with |FCC| > 0.01 were classified as production bottlenecks.

2.11 Multi-target generalization

The pipeline (abbreviated to 150 enumerated reactions per target) was applied without modification to L-malate (EX_mal__L_e) and acetate (EX_ac_e) under the same aerobic glucose conditions, to assess cross-target generalization.

2.12 Literature comparison

Five literature-reported interventions for succinate overproduction were benchmarked against ML predictions:

- ATPS4rpp KO — Mienda & Shamsir, 2016 [9]
- GND KO — Mienda et al., 2016 [10]
- pflB + ldhA double KO — Liang et al., 2007 [11]
- PPC overexpression — Millard et al., 1996 [12]
- PTAr KO — FastKnock, 2023 [4]

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3. Results

Figure 1 presents the 12-panel summary figure (results_summary.png) covering ranking distributions, FBA validation, feature importance, Pareto front, robustness, and multi-target performance.

3.1 Wild-type baseline

Under aerobic glucose minimal medium, the iJO1366 model yields a WT succinate exchange flux of ~0.0001 mmol/gDW/h, consistent with the well-established near-zero aerobic succinate secretion in *E. coli*. WT growth was 0.982 h^{-1}.

3.2 Ablation study

Table 1. Ablation study: 5-fold GroupKFold Spearman ρ.

Model	CV Spearman ρ
GBR-only	**0.557**
Ensemble (GBR + pairwise)	0.358
Pairwise-only	0.159

The pairwise logistic ranker performed poorly in isolation (ρ = 0.159). Including it in an ensemble degraded GBR performance from 0.557 to 0.358. All downstream analyses use GBR-only.

3.3 Holdout generalization

Table 2. Holdout generalization test (70/30 reaction-group split).

Metric	Value
Training reactions / samples	210 / 630
Test reactions / samples	90 / 270
Holdout Spearman ρ	**0.602**
Top-K precision	0.481
Top-5 precision	0.20 (1/5)
Correctly identified top-5 reaction	TKT1 (ko)

The holdout Spearman ρ of 0.602 confirms meaningful generalization to reaction groups absent from training. Top-5 precision of 0.20 indicates that the model provides reliable coarse-grained ranking but is not precise enough to pinpoint the single best intervention in completely novel reaction families without experimental follow-up.

3.4 Top single interventions

Table 3. Top 10 FBA-validated single interventions, sorted by production flux.

Rank	Reaction	Type	Flux (mmol/gDW/h)	Growth (h^{-1})	Composite
1	ATPS4rpp	KO	**16.38**	0.402	0.919
2	O2tpp	KO	15.95	0.242	0.872
3	O2tex	KO	15.95	0.242	0.872
4	ATPM	KO	15.59	0.995	0.968
5	SUCDi	KO	15.44	0.957	0.954
6	G6PDH2r	KO	15.40	0.976	0.955
7	GND	KO	15.40	0.976	0.955
8	PGL	KO	15.40	0.976	0.955
9	RPI	KO	15.40	0.334	0.857
10	NTP1	down	15.39	0.982	0.955

ATPS4rpp KO produces 16.38 mmol/gDW/h succinate — a greater than 100-fold increase over WT — while retaining 41% of WT growth. The disruption of the F_0F_1-ATP synthase forces the cell to reroute carbon through succinate as an alternative electron sink.

3.5 Feature importance

The GBR assigns highest predictive importance to:

1. Intervention type encoding (~40%) — KO vs OE vs downregulation dominates the response
2. FVA minimum flux (fva_min, ~25%) — reactions with constrained lower bounds are harder to remove without metabolic rerouting

These are biologically interpretable: the sign of the perturbation and the degree to which a reaction is metabolically obligatory are the strongest predictors of production response.

3.6 Literature comparison

Table 4. ML predictions vs literature-reported interventions (all originally characterized under anaerobic conditions).

Intervention	ML Rank	FBA Flux (mmol/gDW/h)	Reference
ATPS4rpp KO	**1**	16.38	Mienda & Shamsir, 2016 [9]
GND KO	**9**	15.40	Mienda et al., 2016 [10]
pflB + ldhA double KO	n/a	15.39	Liang et al., 2007 [11]
PPC OE	1329	15.39	Millard et al., 1996 [12]
PTAr KO	6242	15.39	FastKnock, 2023 [4]

All five literature interventions were originally proposed or validated under anaerobic glucose fermentation, whereas our simulations are aerobic. This context difference explains the divergent rankings: ATPS4rpp KO and GND KO disrupt ATP synthesis and the pentose phosphate pathway respectively — pathways relevant under both aerobic and anaerobic energy metabolism — and are correctly recovered in the top 10. PPC overexpression and PTAr KO act primarily by increasing reductive TCA flux, a mechanism specific to anaerobic succinate production, and therefore show negligible aerobic effect (flux ≈ WT baseline at 15.39 mmol/gDW/h). Notably, Mienda & Shamsir (2016) report ATPS4rpp KO as a computational prediction rather than an experimentally confirmed deletion, strengthening the parallel between our approach and prior in silico work.

3.7 Double interventions

Table 5. Top ML-predicted double interventions validated with FBA.

Intervention	Flux (mmol/gDW/h)	Growth (h^{-1})	Composite
ATPM KO + O2tex KO	**16.80**	0.276	0.921
ATPM KO + O2tpp KO	16.80	0.276	0.921
ATPM KO + ATPS4rpp KO	16.40	0.437	0.925
ATPM KO + SUCDi KO	15.63	0.971	0.966
ATPM KO + PGL KO	15.61	0.988	0.968

The best ML-identified double intervention (ATPM KO + O2tex KO) achieves 16.80 mmol/gDW/h — a 9% improvement over the best single knockout. ML-guided search identified 4 of 25 validated pairs with composite score > 0.95, compared to 0 of 25 for random double selection. The best flux found by random pairs was 15.39 mmol/gDW/h, the same as the unenumerated single baseline.

3.8 Flux control coefficient analysis

SUCCtex (FCC = 0.947) is identified as the dominant production bottleneck — the succinate transport reaction from cytoplasm to periplasm controls the majority of flux toward secretion. PGK and GAPD (FCC ≈ 0.084 each) are secondary glycolytic bottlenecks.

3.9 Monte Carlo robustness

Table 6. Robustness of top interventions under Gaussian parameter noise (σ = 20%, 15 trials).

Intervention	Mean Flux	Std	Robustness Score
ATPS4rpp KO	17.07	2.47	**14.60**
ATPM KO	16.38	2.42	13.96
SUCDi KO	15.86	2.97	12.89
O2tex KO	14.81	3.01	11.80
G6PDH2r KO	15.34	3.61	11.74

ATPS4rpp KO maintains the highest mean production flux and the highest robustness score across all 15 stochastic parameter perturbations, confirming it as the most reliable engineering target.

3.10 Multi-target generalization

Table 7. Pipeline performance across three production targets.

Target	Reaction	CV ρ	Top-K precision	Best single flux (mmol/gDW/h)	Best KO
Succinate	EX_succ_e	0.557	0.472	16.38	ATPS4rpp
L-malate	EX_mal__L_e	0.445	0.333	16.79	ATPM
Acetate	EX_ac_e	0.491	0.367	27.80	ATPS4rpp

The pipeline generalizes across all three targets without retraining, maintaining Spearman correlations of ρ = 0.445–0.557 and identifying ATPS4rpp KO as a consistently high-ranking target for both succinate and acetate production.

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4. Discussion

RéseauFlux demonstrates that a gradient boosting regressor trained on approximately 300 FBA-enumerated interventions can meaningfully rank the full intervention space of a 2,583-reaction genome-scale model. The holdout Spearman ρ of 0.602 on reaction groups absent from training confirms genuine generalization, addressing a weakness common in ML-based strain design methods that conflate cross-validation performance with true out-of-distribution generalization.

The ablation study reveals an important negative result: a pairwise logistic ranker degrades performance when ensembled with GBR (ρ drops from 0.557 to 0.358). The pairwise ranker's poor standalone performance (ρ = 0.159) suggests that pairwise flux difference signals are too noisy at genome scale for this approach to be beneficial. This is a methodologically useful finding — ensemble strategies are not universally beneficial, and ablation studies should be a standard component of published ML pipelines for metabolic engineering.

The Top-5 precision of 0.20 on the holdout set reflects a genuine limitation: while the model provides useful coarse-grained ranking, it does not reliably recover the single best interventions for completely novel reaction families. We recommend RéseauFlux as a screening tool to reduce the experimental search space from thousands of candidates to a shortlist of 50–100 for targeted FBA validation or wet-lab characterization, rather than as a direct replacement for exhaustive simulation.

The ML-guided discovery of ATPM KO + O2tex KO (16.80 mmol/gDW/h) illustrates the practical value: exhaustive FBA evaluation of all pairwise interventions in iJO1366 would require approximately 7 million simulations, whereas RéseauFlux reduces this to 25 targeted validations with measurably better outcomes than random selection (4/25 vs 0/25 above the high-composite threshold).

All five benchmarked literature interventions were characterized under anaerobic conditions, while our simulations are aerobic. The aerobic/anaerobic context gap explains why reductive TCA strategies (PPC OE, PTAr KO) rank poorly, and why energy-metabolism disruptions (ATPS4rpp KO, GND KO) rank well regardless of oxygen availability. Future work should extend the pipeline to anaerobic conditions to enable direct comparison with experimental succinate titers.

Future directions include: incorporating graph neural network features over the stoichiometric bipartite graph to improve generalization to structurally distant reactions; adaptive enumeration strategies that iteratively prioritize informative reactions for FBA labeling; and integration with experimental feedback for active learning.

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5. Conclusion

RéseauFlux provides an open, reproducible, and computationally efficient framework for ML-guided metabolic intervention ranking in genome-scale models. On aerobic succinate production in *E. coli* iJO1366, the pipeline correctly prioritizes ATPS4rpp KO as the top candidate — consistent with prior computational predictions in the succinate engineering literature — identifies a novel double intervention that outperforms all validated singles, and generalizes meaningfully to unseen reaction groups and alternative production targets. All code and outputs are publicly available.

Data and Code Availability

All code is available at github.com/Daxlia/ReseauFlux under a Modified MIT License with Citation Requirement. The iJO1366 genome-scale model is publicly available through the BiGG Models database and is loaded automatically by COBRApy. All result files (CSV tables and summary figure) can be reproduced within expected stochastic variation by executing `python metabolic_ml_pipeline.py` from the repository root.

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Reproducibility Notes

Reported results correspond to the author’s original experimental runs. Due to differences in random seed configuration, subsequent executions may yield moderately different rankings, validation scores, and numerical outputs despite identical pipeline settings.

AI Disclosure

This manuscript was prepared with the assistance of Claude (claude-sonnet-4-6, Anthropic, PBC). The AI was used for: Python code development and debugging, manuscript drafting and revision, verification of technical claims against the codebase, and literature cross-referencing. All scientific decisions, experimental design, result interpretation, and final content responsibility remain solely with the author.

Conflict of Interest

The author declares no competing interests.

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